

WEEK-4

Write a program to solve XOR problem using backpropagation algorithm in multilayer neural networks.

```
w11,w12=1.2,0.1
w21,w22=0.3,0.4
w13,w23=0.1,0.2
x1 = [0,0,1,1]
x2 = [0,1,0,1]
y = [0,1,1,0]
import math
lr = 0.5
epochs = 3
def sigmoid(x):
    return 1/(1+math.exp(-x))
for ep in range(epochs):
    print("Epoch {}".format(ep+1))
    for i in range(4):
        #Forward pass
        h1_in = w11*x1[i]+ w21*x2[i]
        h1_out = sigmoid(h1_in)
        h2_in = w12*x1[i]+w22*x2[i]
        h2_out = sigmoid(h2_in)
        o_in = w13*h1_out+w23*h2_out
        o = sigmoid(o_in)
```

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```
error = y[i]-o
print(f"Sample:{i+1} Error:{error}")

#Backward pass
d3 = o*(1-o)*error
d2 = h2_out*(1-h2_out)*d3*w23
d1 = h1_out*(1-h1_out)*d3*w13

#weight updation
w13+=(lr*d3*h1_out)
w23+=(lr*d3*h2_out)
w12+=(lr*d2*x1[i])
w11+=(lr*d1*x1[i])
w21+=(lr*d1*x2[i])
w22+=(lr*d2*x2[i])

print(f"Weights: w11={w11:.3f}, w12={w12:.3f}, w21={w21:.3f}, w22={w22:.3f}, w13={w13:.3f},
w23={w23:.3f}")
```

OUTPUT:

Sample:1 Error:-0.5374298453437496

Weights: w11=1.200, w12=0.100, w21=0.300, w22=0.400, w13=0.067, w23=0.167

Sample:2 Error:0.4655551678297135

Weights: w11=1.200, w12=0.100, w21=0.301, w22=0.402, w13=0.100, w23=0.201

Sample:3 Error:0.45452184567765197

Weights: w11=1.201, w12=0.103, w21=0.301, w22=0.402, w13=0.143, w23=0.231

Sample:4 Error:-0.5648996979715675

Weights: w11=1.200, w12=0.099, w21=0.299, w22=0.399, w13=0.086, w23=0.188

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Epoch 2:

Sample:1 Error:-0.5341904202588642

Weights: w11=1.200, w12=0.099, w21=0.299, w22=0.399, w13=0.053, w23=0.154

Sample:2 Error:0.46932062874229785

Weights: w11=1.200, w12=0.099, w21=0.300, w22=0.401, w13=0.087, w23=0.189

Sample:3 Error:0.45860057815700694

Weights: w11=1.200, w12=0.102, w21=0.300, w22=0.401, w13=0.130, w23=0.219

Sample:4 Error:-0.5605117643902519

Weights: w11=1.199, w12=0.098, w21=0.299, w22=0.397, w13=0.074, w23=0.176

Epoch 3:

Sample:1 Error:-0.5312323426581409

Weights: w11=1.199, w12=0.098, w21=0.299, w22=0.397, w13=0.041,
w23=0.143

Sample:2 Error:0.4727568255775332

Weights: w11=1.199, w12=0.098, w21=0.299, w22=0.399, w13=0.075,
w23=0.178

Sample:3 Error:0.4623228892742921

Weights: w11=1.200, w12=0.101, w21=0.299, w22=0.399, w13=0.119,
w23=0.208

Sample:4 Error:-0.556503005855365

Weights:w11=1.199,w12=0.097,w21=0.298,w22=0.396, w13=0.063,
w23=0.166

Optional: Epoch 1: Sample:1 Error:-

0.5374298453437496

Sample:2

Error:0.4655551678297135

Sample:3

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Error:0.45452184567765197

Sample:4 Error:-

0.5648996979715675

Epoch 10001:

Sample:1 Error:-

0.029974038639471665 Sample:2

Error:0.5003712854270218

Sample:3

Error:0.028601876228930623

Sample:4 Error:-

0.5009350158841845

Epoch 20001:

Sample:1 Error:-

0.01952740012025413 Sample:2

Error:0.5001824865524209

Sample:3

Error:0.01827318660498045

Sample:4 Error:-

0.5003808368580002

Epoch 30001:

Sample:1 Error:-

0.015507232163392386 Sample:2

Error:0.5001213927611297

Sample:3

Error:0.014385984063787882

Sample:4 Error:-

0.5002352020933362

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